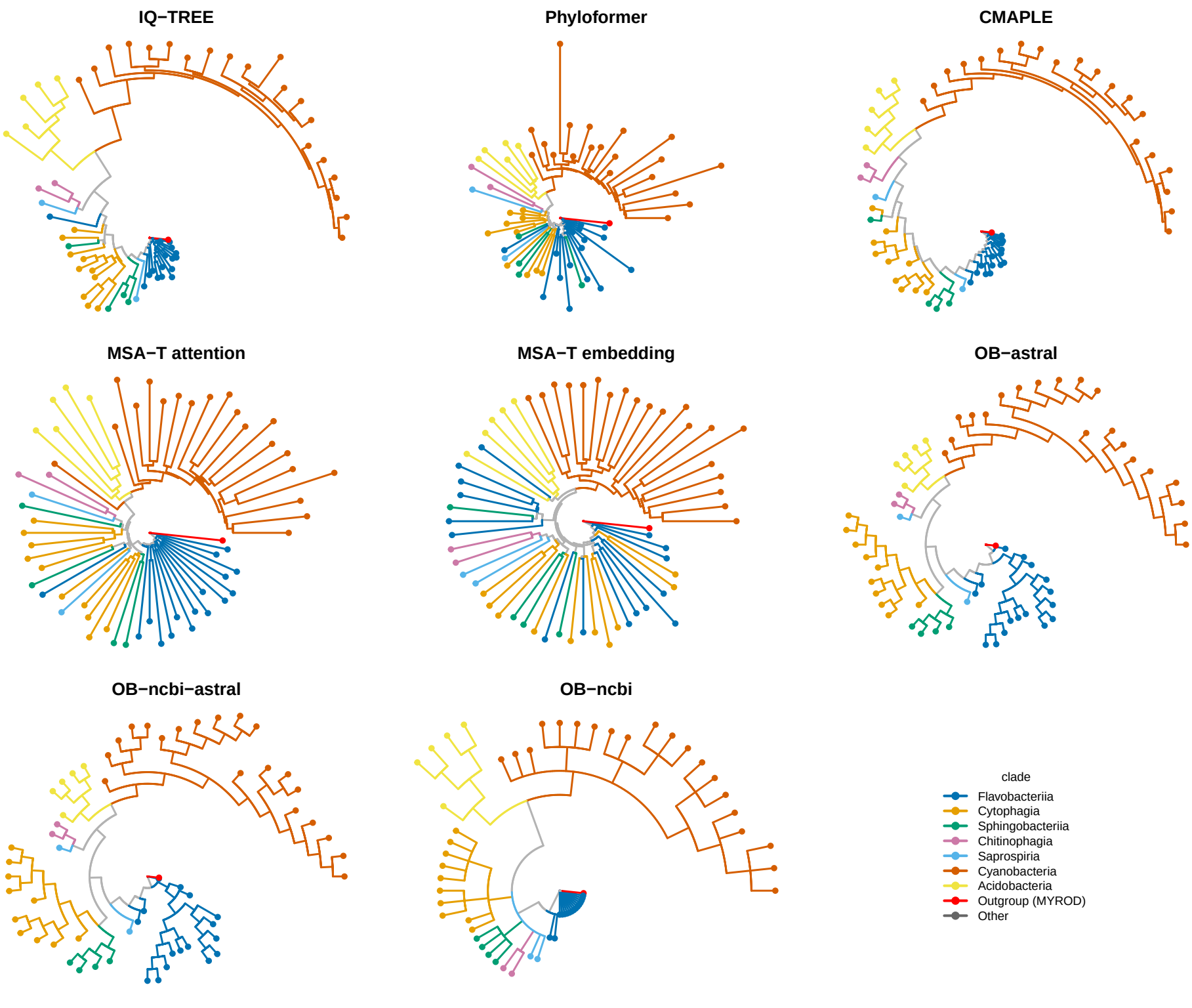
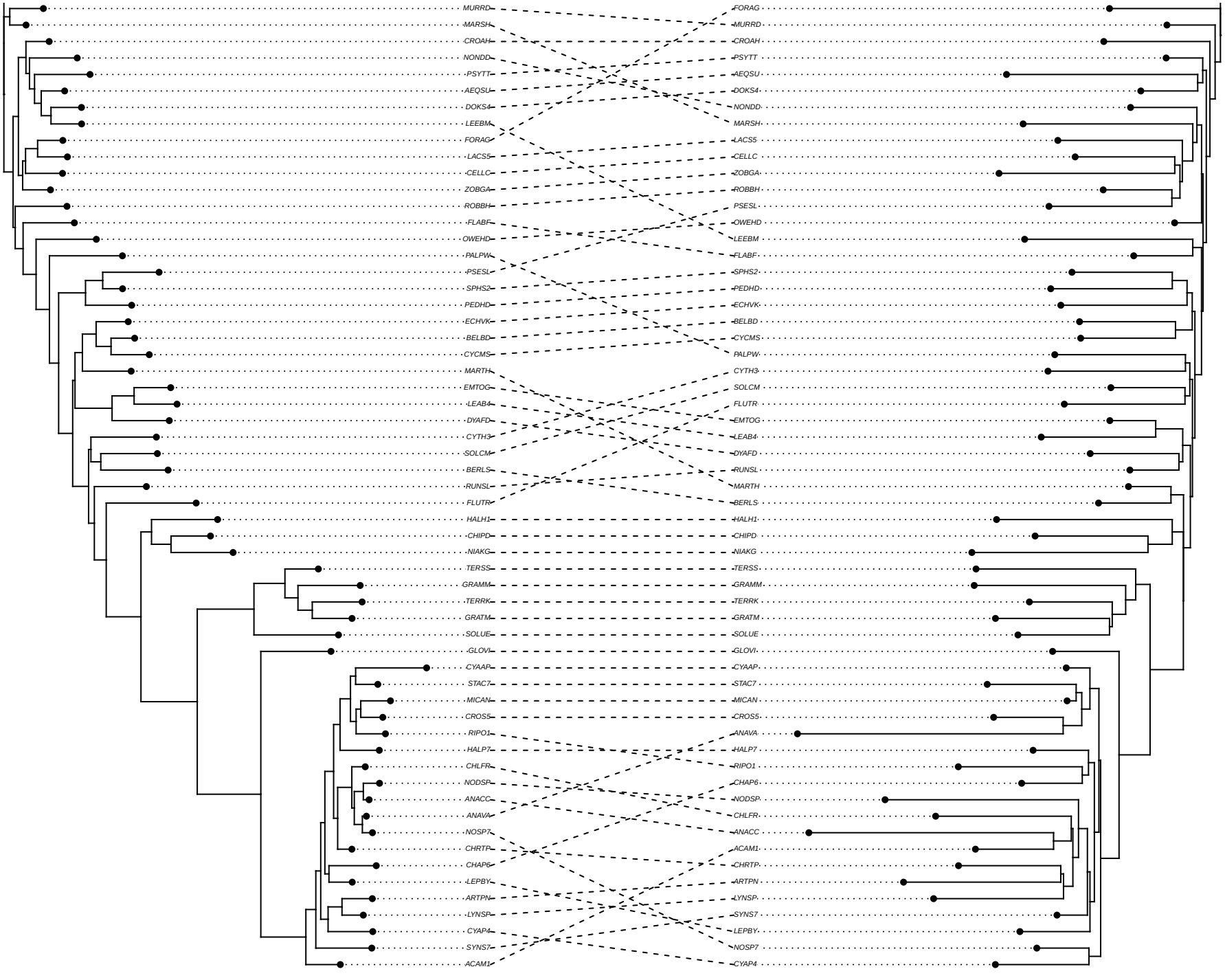
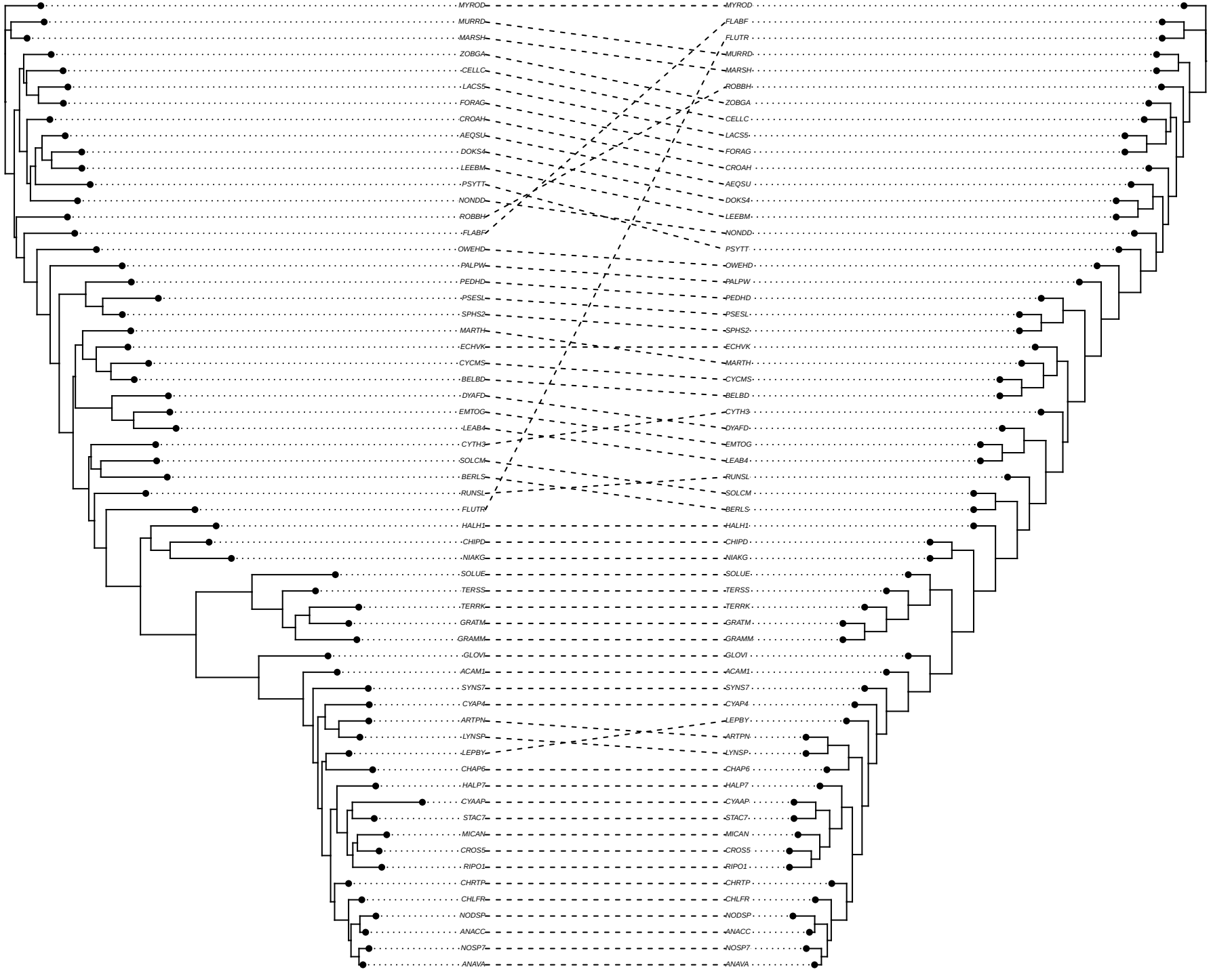
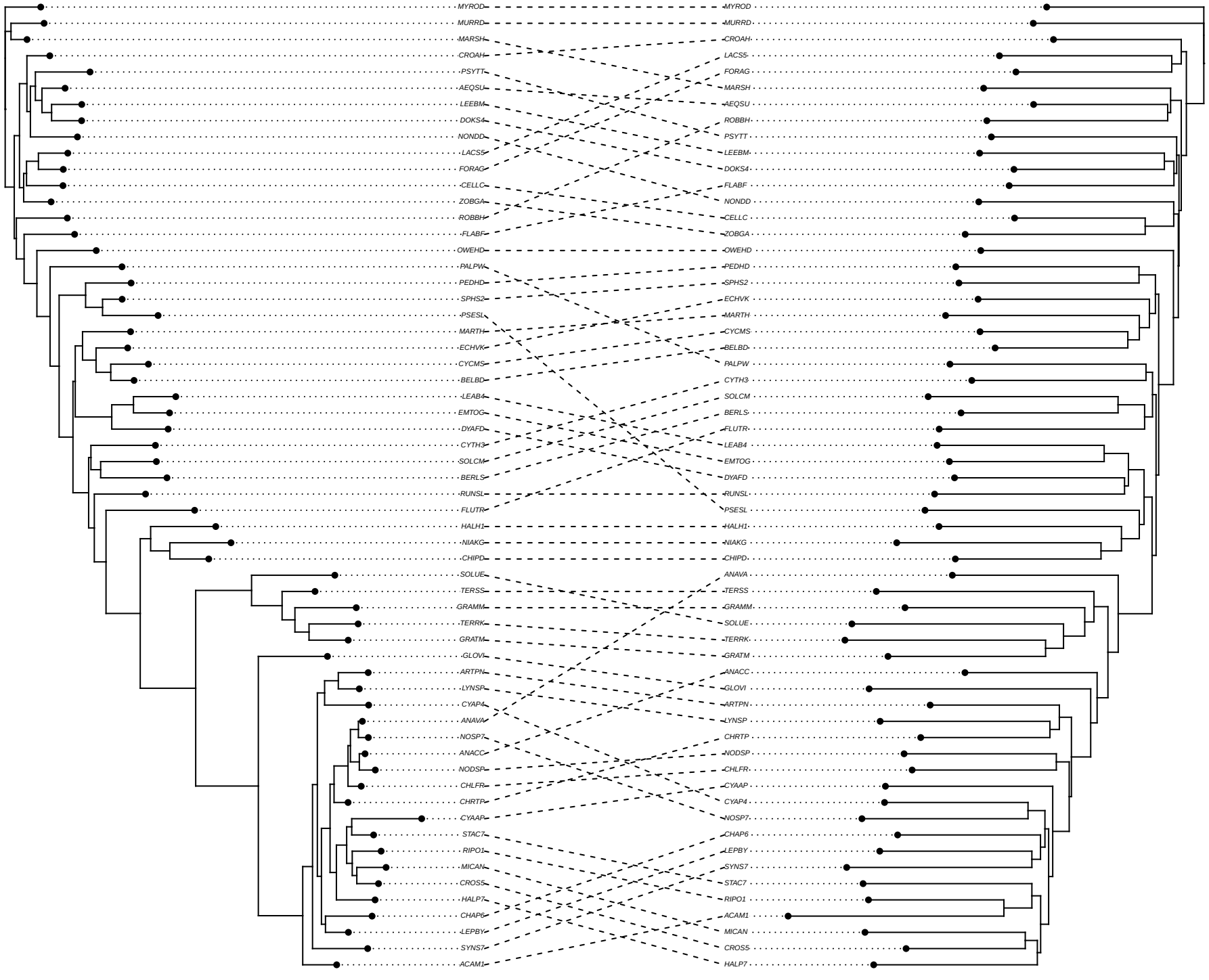
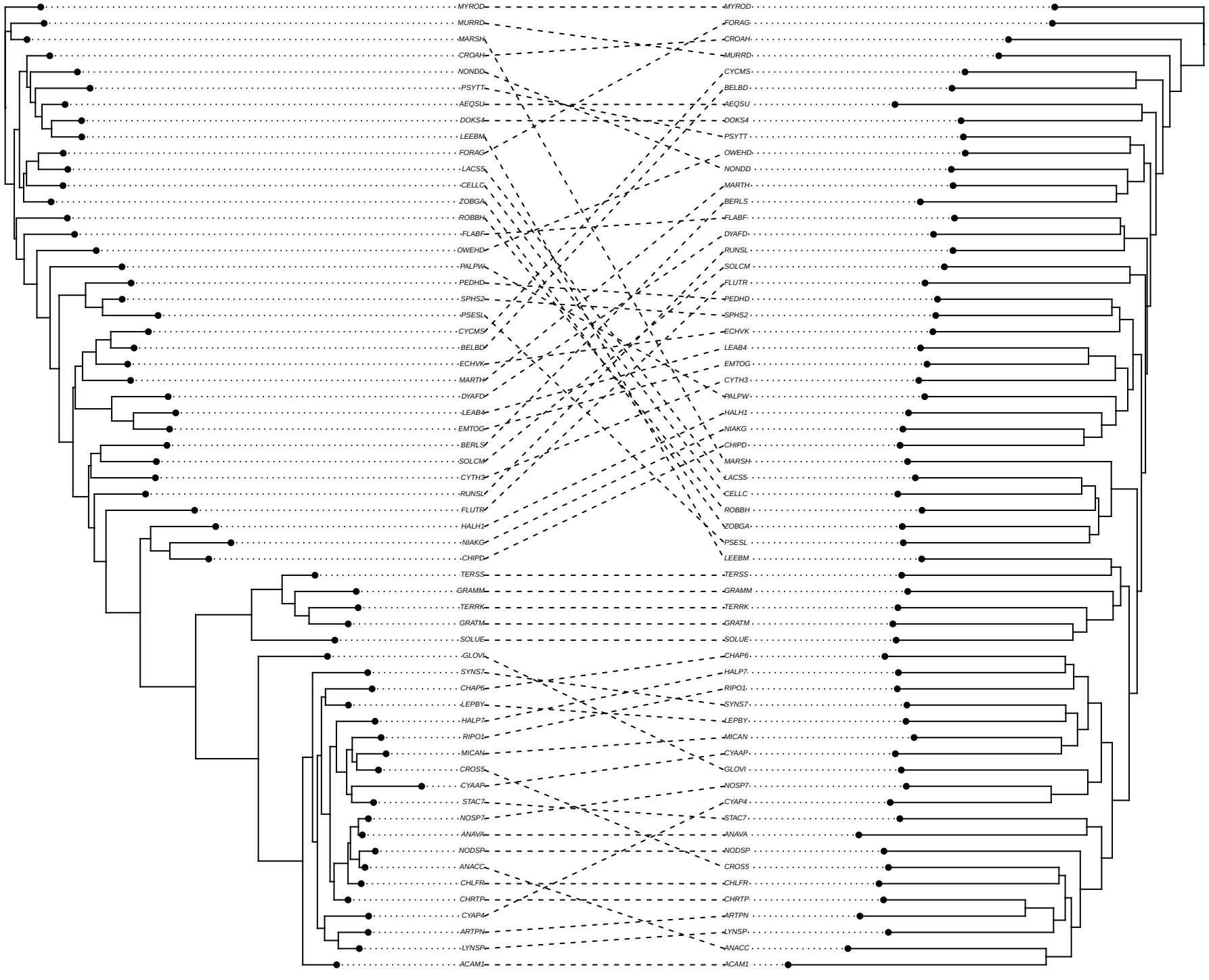

IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
OB−astral
OB−ncbi−astral
OB−ncbi
clade
Flavobacteriia
Cytophagia
Sphingobacteriia
Chitinophagia
Saprospiria
Cyanobacteria
Acidobacteria
Outgroup (MYROD)
Other


IQ−TREE vs Phylolformer
MURRD
MARSH
CROAH
NONDD
PSYTT
AEQSU
DOKS4
LEEBM
FORAG
LACS5
CELLC
ZOBGA
ROBBH
FLABF
OWEHD
PALPW
PSESL
SPHS2
PEDHD
ECHVK
BELBD
CYCMS
MARTH
EMTOG
LEAB4
DYAFD
CYTH3
SOLCM
BERLS
RUNSL
FLUTR
HALH1
CHIPD
NIAKG
TERSS
GRAMM
TERRK
GRATM
SOLUE
GLOVI
CYAAP
STAC7
MICAN
CROS5
RIPO1
HALP7
CHLFR
NODSP
ANACC
ANAVA
NOSP7
CHRTP
CHAP6
LEPBY
ARTPN
LYNSP
CYAP4
SYNS7
ACAM1
FORAG
MURRD
CROAH
PSYTT
AEQSU
DOKS4
NONDD
MARSH
LACS5
CELLC
ZOBGA
ROBBH
PSESL
OWEHD
LEEBM
FLABF
SPHS2
PEDHD
ECHVK
BELBD
CYCMS
PALPW
CYTH3
SOLCM
FLUTR
EMTOG
LEAB4
DYAFD
RUNSL
MARTH
BERLS
HALH1
CHIPD
NIAKG
TERSS
GRAMM
TERRK
GRATM
SOLUE
GLOVI
CYAAP
STAC7
MICAN
CROS5
ANAVA
HALP7
RIPO1
CHAP6
NODSP
CHLFR
ANACC
ACAM1
CHRTP
ARTPN
LYNSP
SYNS7
LEPBY
NOSP7
CYAP4

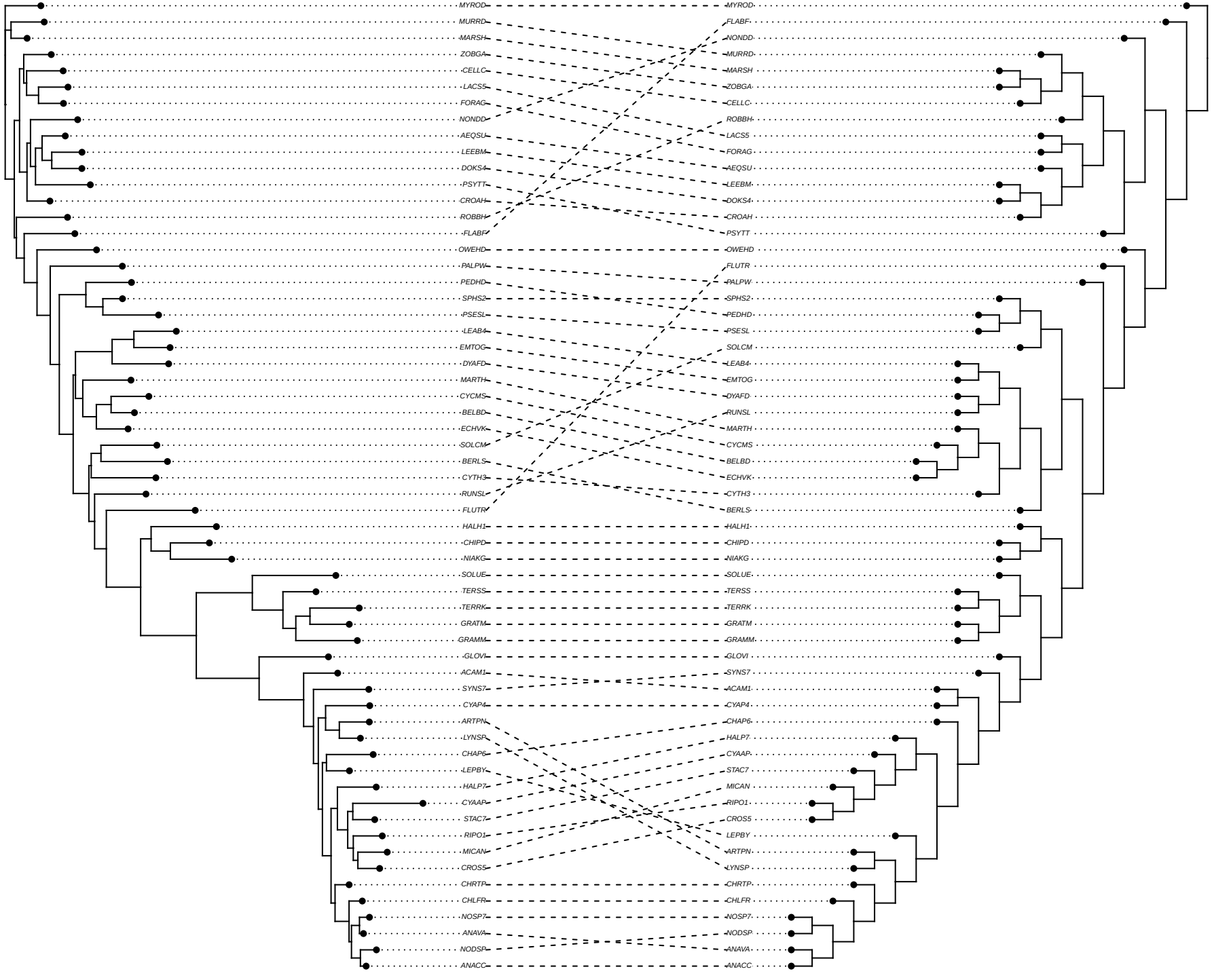
IQ-TREE vs CMAPLE



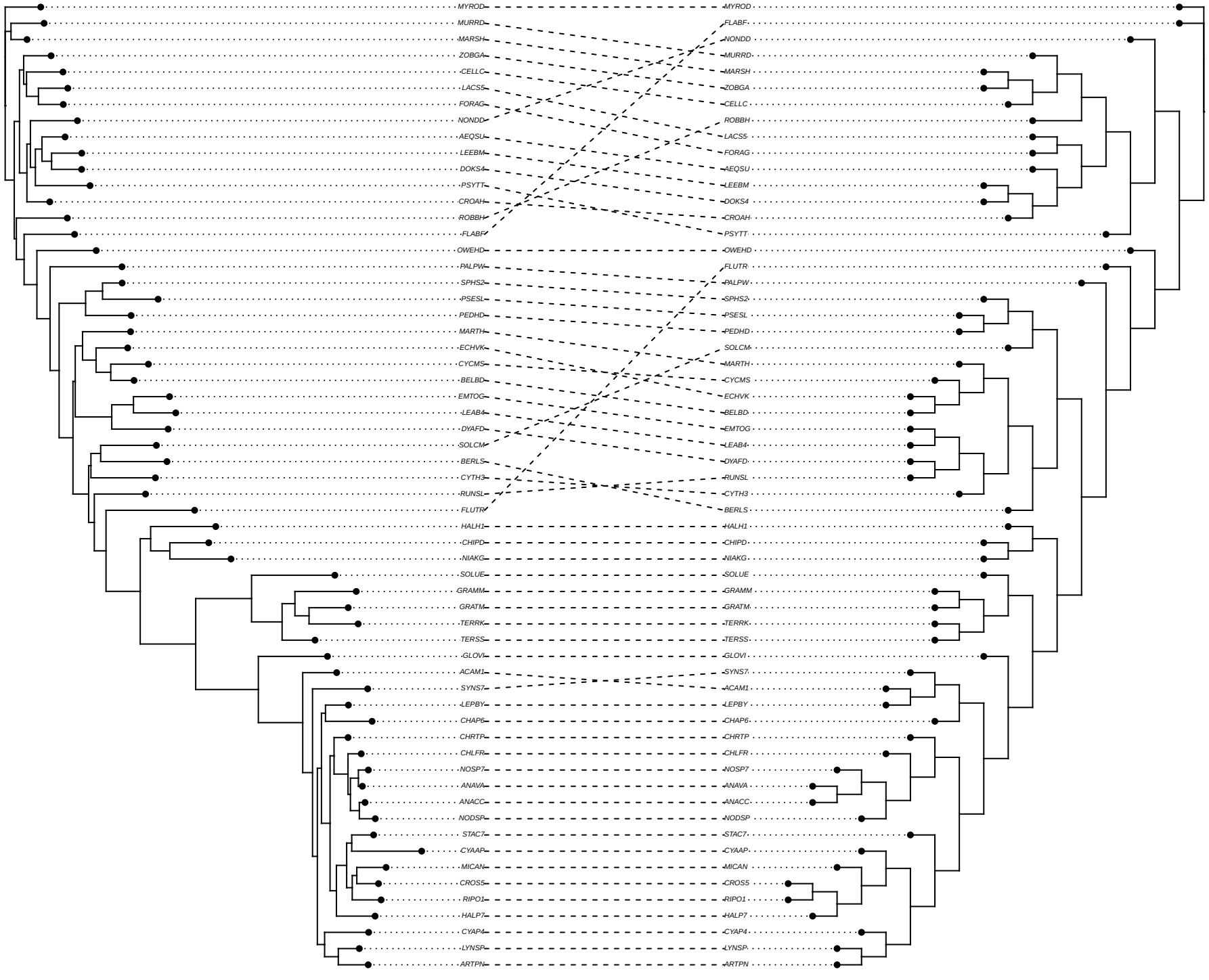

IQ−TREE vs MSA−Pattention
MYROD
MURRD
MARSH
CROAH
PSYTT
AEQSU
LEEBM
DOKS4
NONDD
LACS5
FORAG
CELLC
ZOBGA
ROBBH
FLABF
OWEHD
PALPW
PEDHD
SPHS2
PSESL
MARTH
ECHVK
CYCMS
BELBD
LEAB4
EMTOG
DYAFD
CYTH3
SOLCM
BERLS
RUNSL
FLUTR
HALH1
NIAKG
CHIPD
SOLUE
TERSS
GRAMM
TERRK
GRATM
GLOVI
ARTPN
LYNSP
CYAP4
ANAVA
NOSP7
ANACC
NODSP
CHLFR
CHRTP
CYAAP
STAC7
RIPO1
MICAN
CROS5
HALP7
CHAP6
LEPBY
SYNS7
ACAM1
MYROD
MURRD
CROAH
LACS5
FORAG
MARSH
AEQSU
ROBBH
PSYTT
LEEBM
DOKS4
FLABF
NONDD
CELLC
ZOBGA
OWEHD
PEDHD
SPHS2
ECHVK
MARTH
CYCMS
BELBD
PALPW
CYTH3
SOLCM
BERLS
FLUTR
LEAB4
EMTOG
DYAFD
RUNSL
PSESL
HALH1
NIAKG
CHIPD
ANAVA
TERSS
GRAMM
SOLUE
TERRK
GRATM
ANACC
GLOVI
ARTPN
LYNSP
CHRTP
NODSP
CHLFR
CYAAP
CYAP4
NOSP7
CHAP6
LEPBY
SYNS7
STAC7
RIPO1
ACAM1
MICAN
CROS5
HALP7


IQ−TREE vs MSA−T embedding
MYROD
MURRD
MARSH
CROAH
NONDD
PSYTT
AEQSU
DOKS4
LEEBM
FORAG
LACS5
CELLC
ZOBGA
ROBBH
FLABF
OWEHD
PALPW
PEDHD
SPHS2
PSESL
CYCMS
BELBD
ECHVK
MARTH
DYAFD
LEAB4
EMTOG
BERLS
SOLCM
CYTH3
RUNSL
FLUTR
HALH1
NIAKG
CHIPD
TERSS
GRAMM
TERRK
GRATM
SOLUE
GLOVI
SYNS7
CHAP6
LEPBY
HALP7
RIPO1
MICAN
CROS5
CYAAP
STAC7
NOSP7
ANAVA
NODSP
ANACC
CHLFR
CHRTP
CYAP4
ARTPN
LYNSP
ACAM1
MYROD
FORAG
CROAH
MURRD
CYCMS
BELBD
AEQSU
DOKS4
PSYTT
OWEHD
NONDD
MARTH
BERLS
FLABF
DYAFD
RUNSL
SOLCM
FLUTR
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NIAKG
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ROBBH
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LEEBM
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MICAN
CYAAP
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CROS5
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CHRTP
ARTPN
LYNSP
ANACC
ACAM1

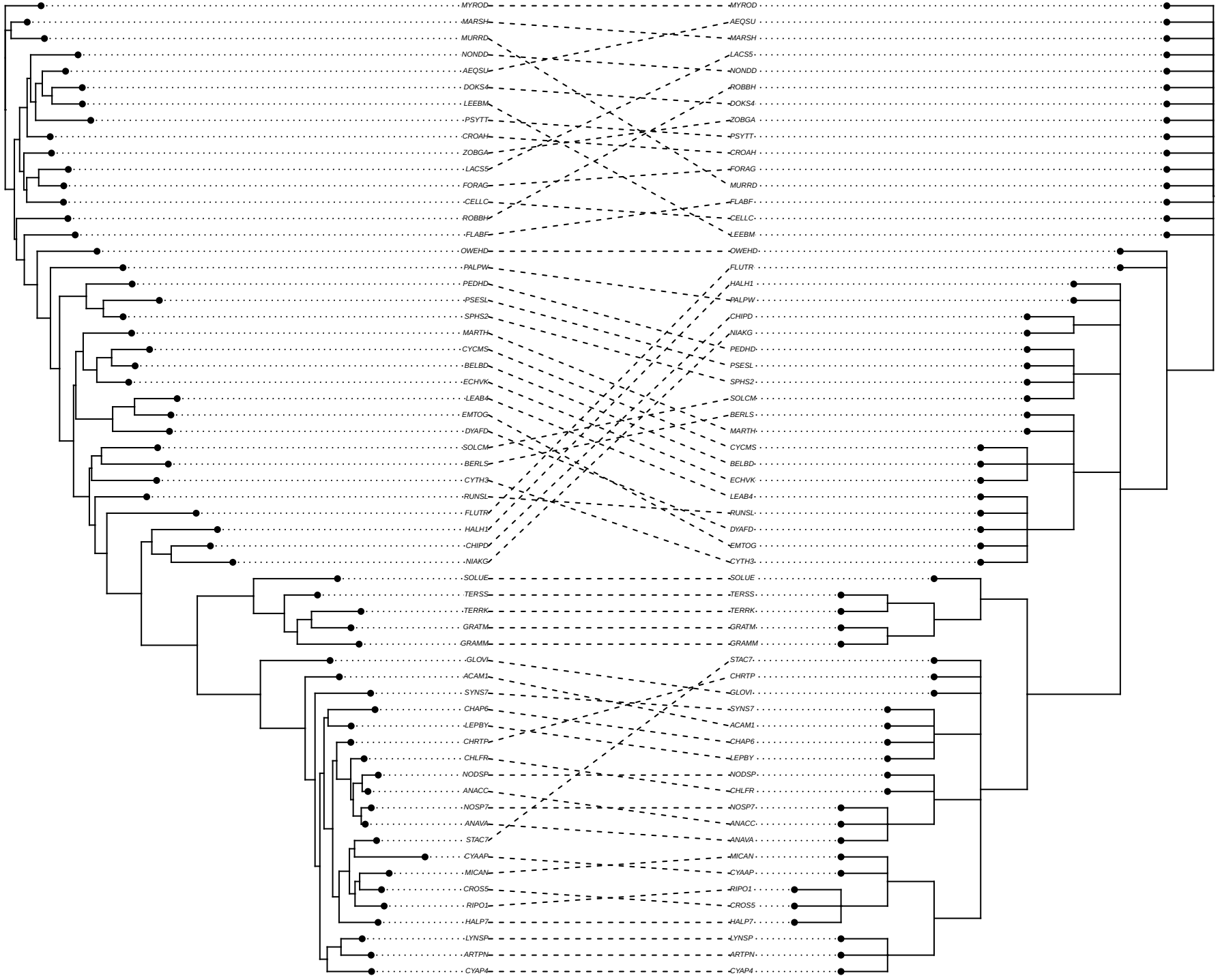
IQ−TREE vs OB−astral

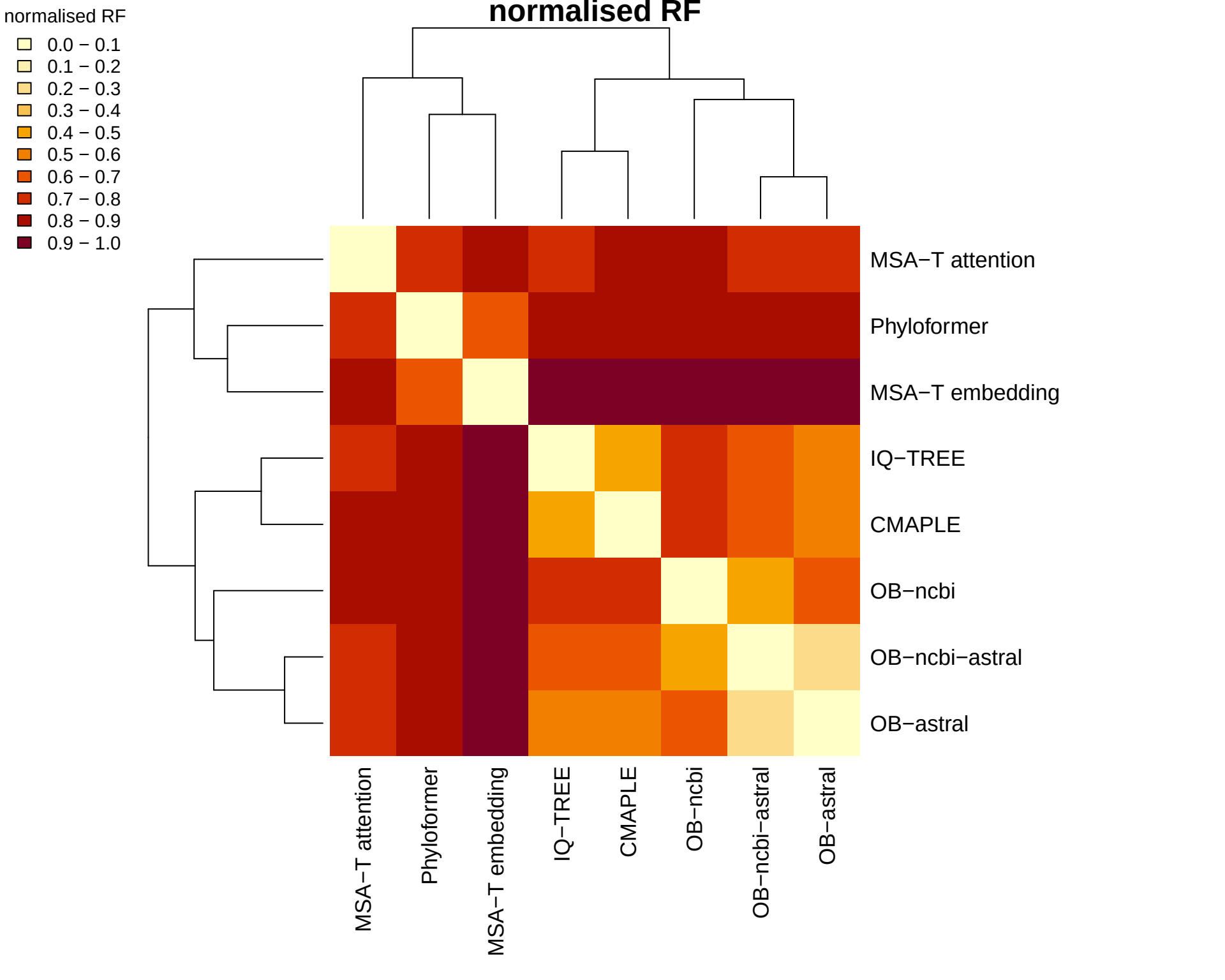


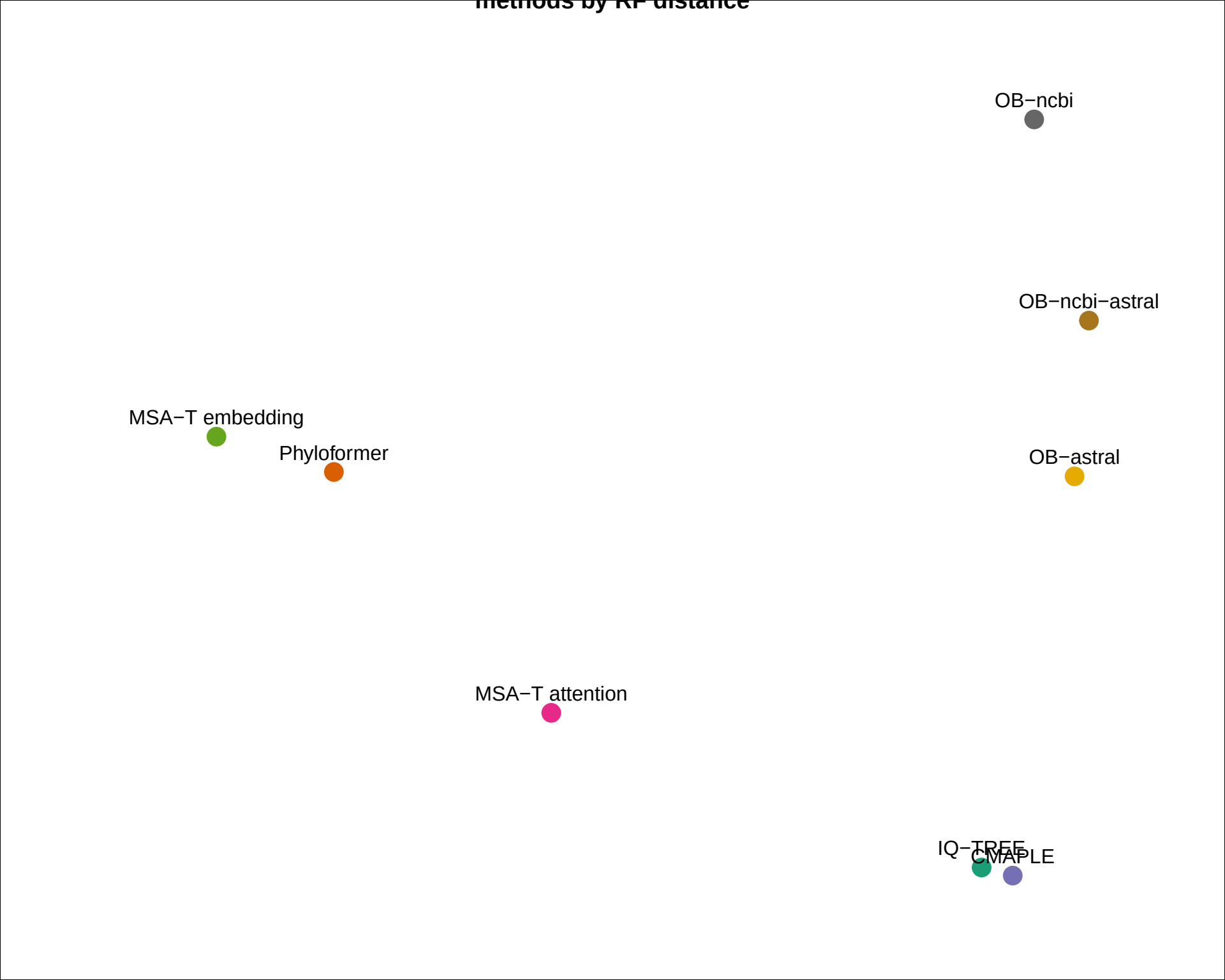
IQ-TREE vs OB-ncbi-astral

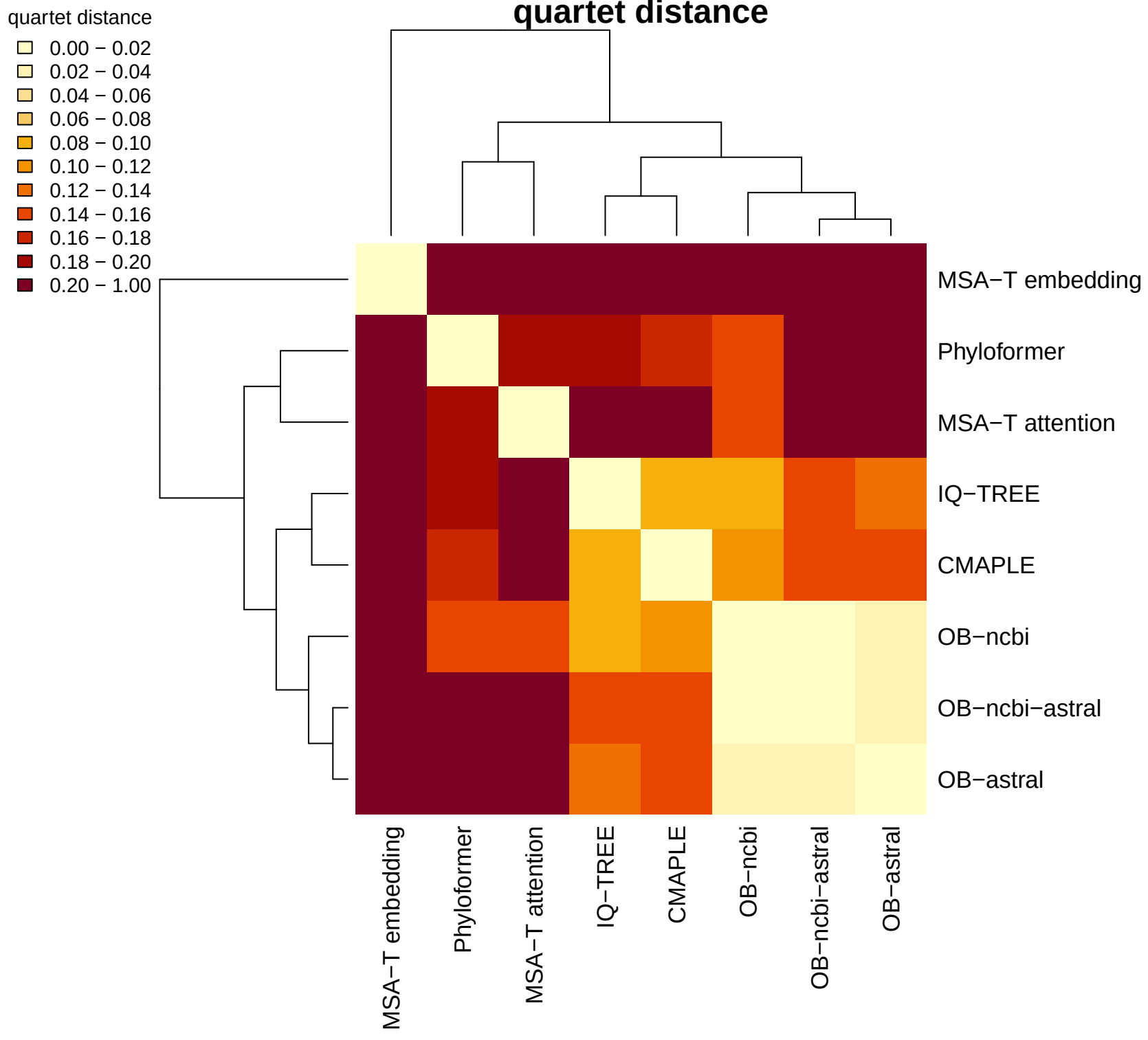


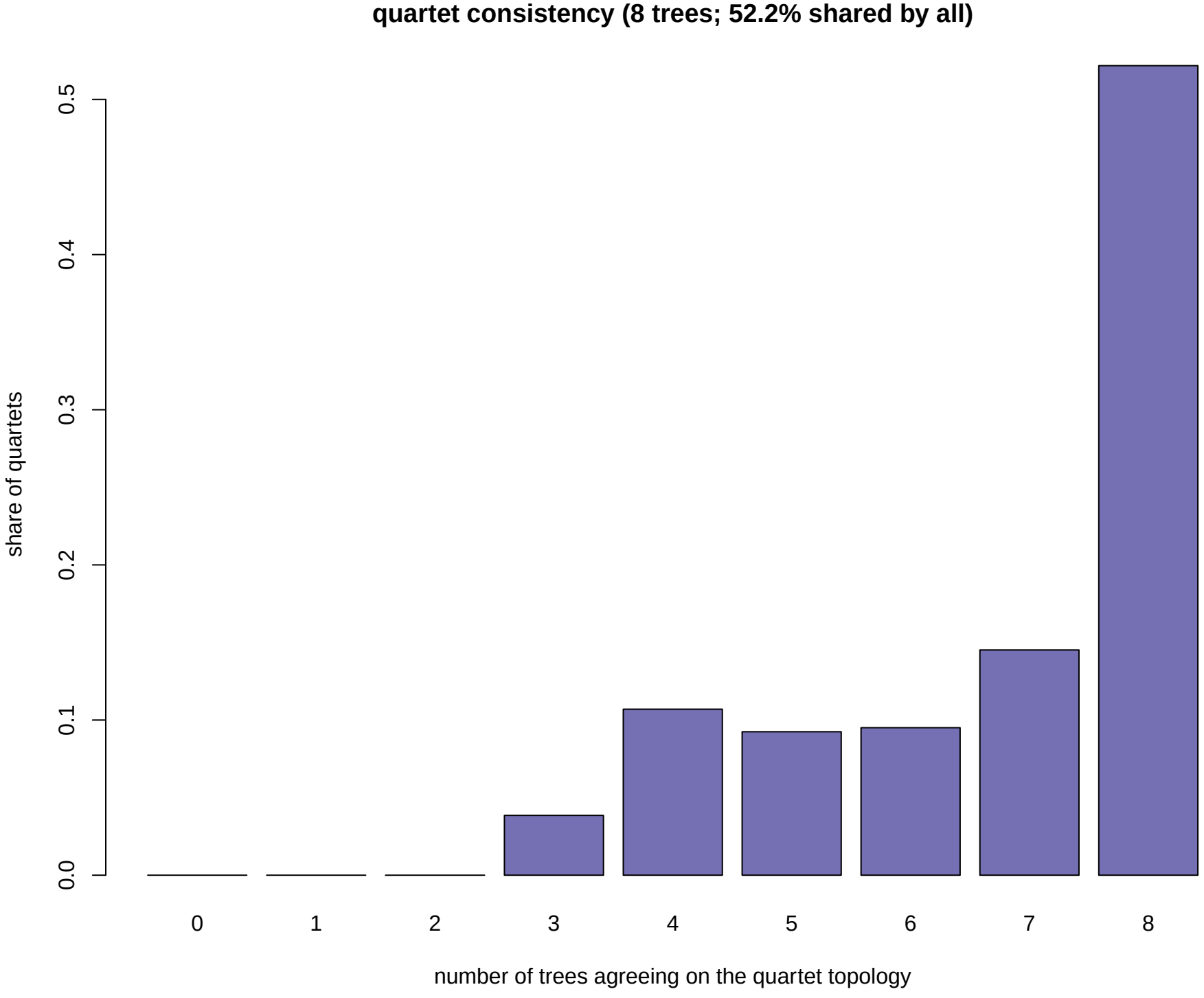
IQ-TREE vs OB-ncbi

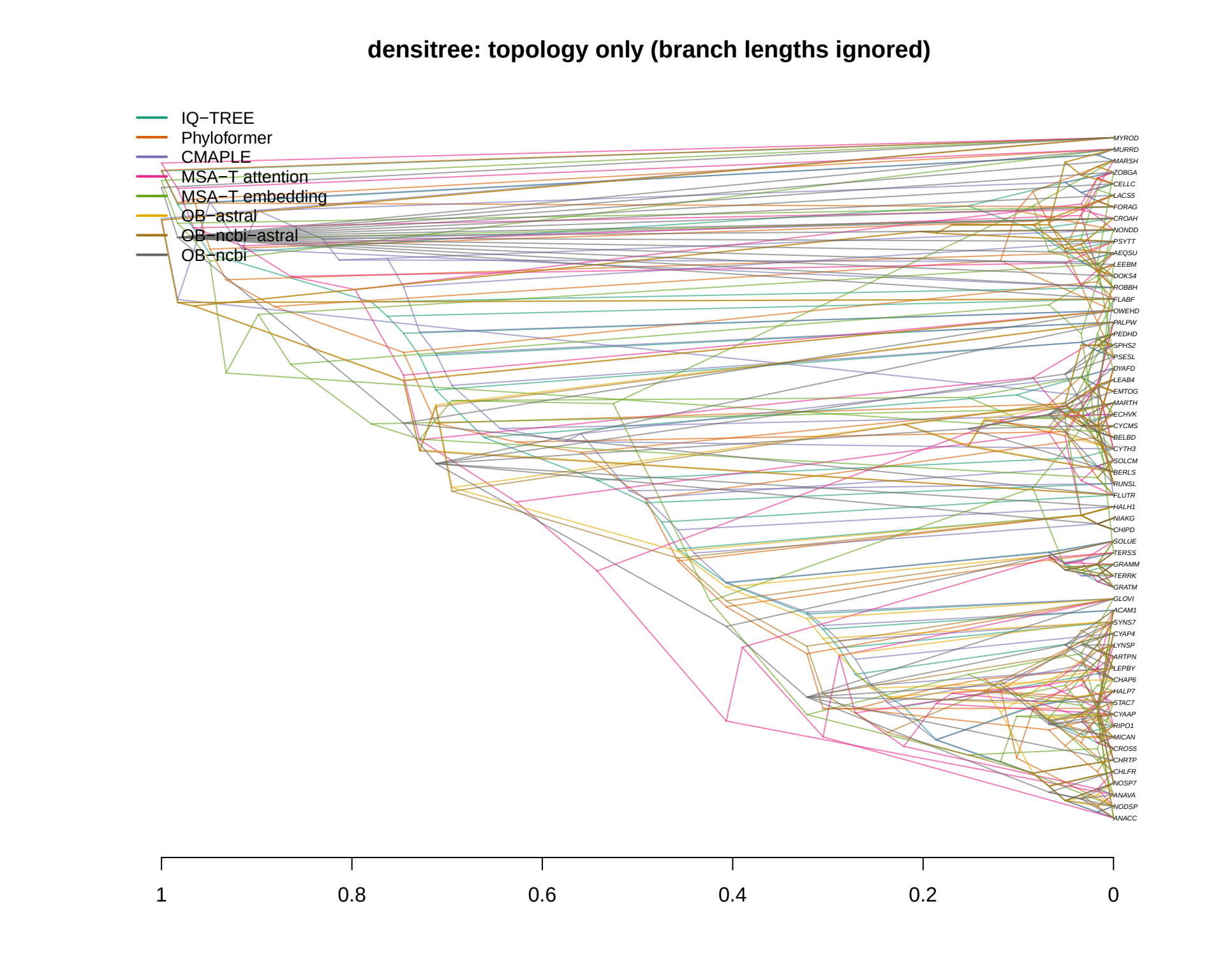


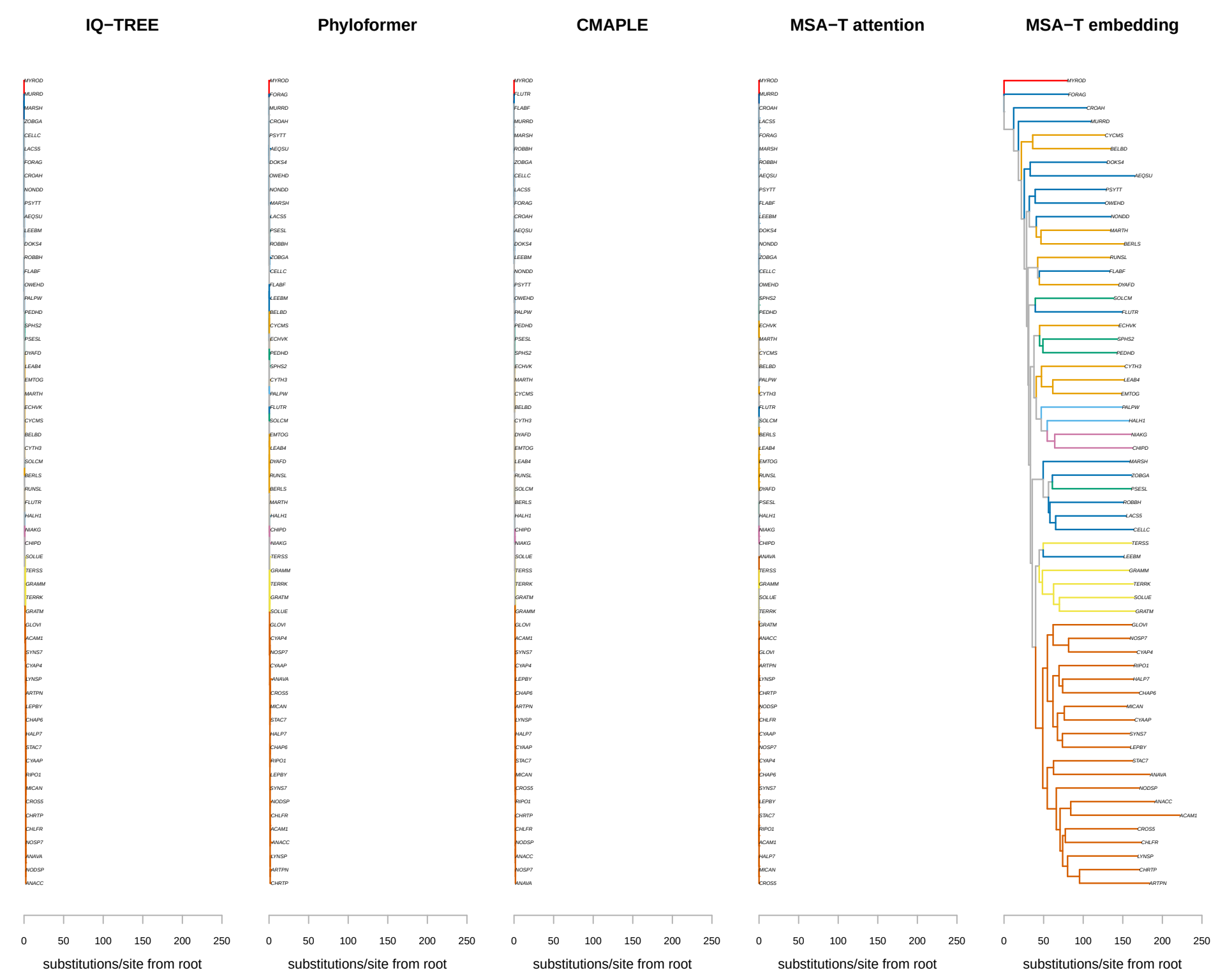

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T attention
Phyloformer
MSA−T embedding
IQ−TREE
CMAPLE
OB−ncbi
OB−ncbi−astral
OB−astral


methods by RF distance
OB−ncbi
OB−ncbi−astral
MSA−T embedding
Phyloformer
OB−astral
MSA−T attention
IQ−TREE
CMAPLE

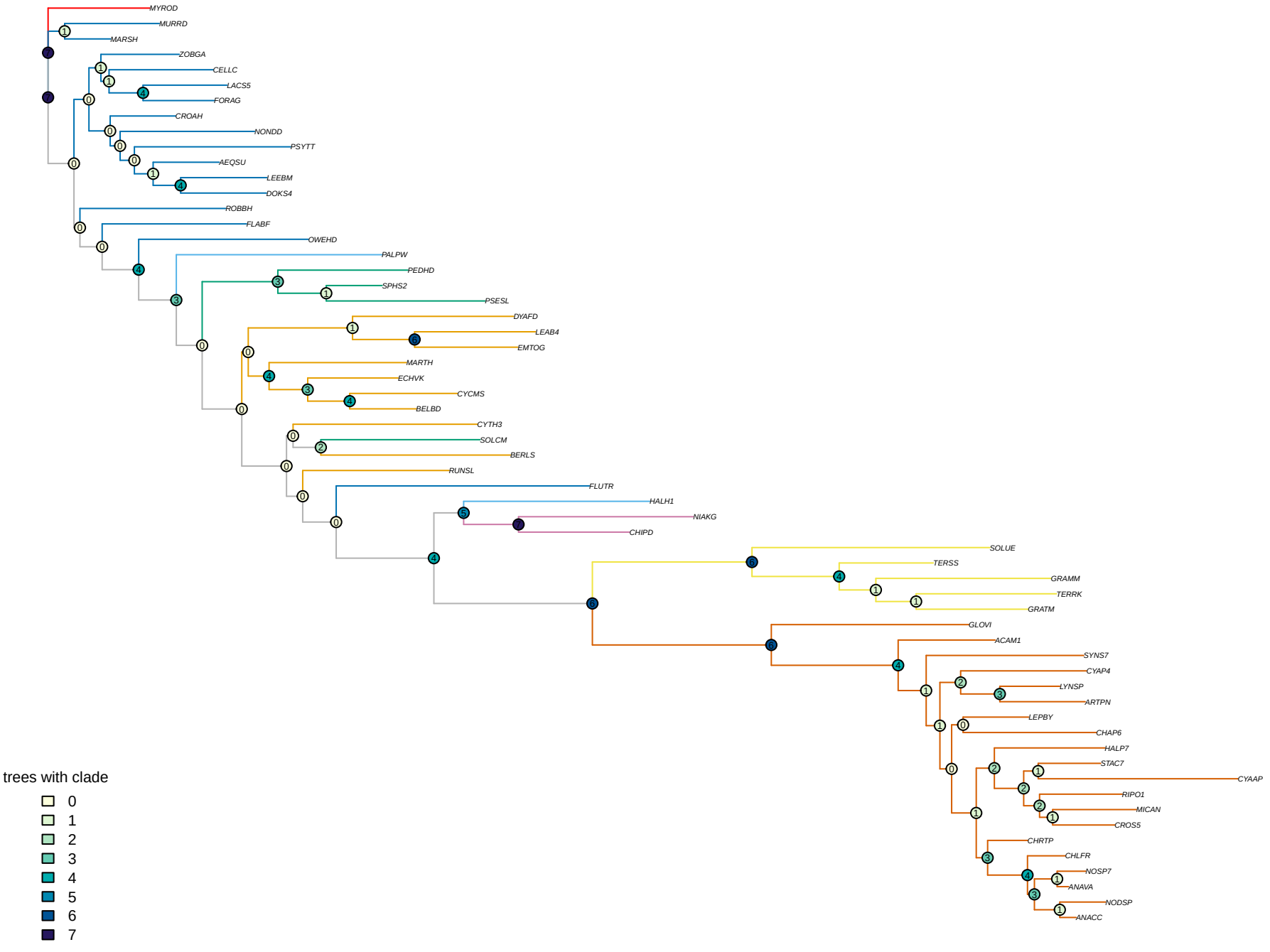

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
MSA−T embedding
Phyloformer
MSA−T attention
IQ−TREE
CMAPLE
OB−ncbi
OB−ncbi−astral
OB−astral
MSA−T embedding
Phyloformer
MSA−T attention
IQ−TREE
CMAPLE
OB−ncbi
OB−ncbi−astral
OB−astral


quartet consistency (8 trees; 52.2% shared by all)
share of quartets
0.0
0.1
0.2
0.3
0.4
0.5
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

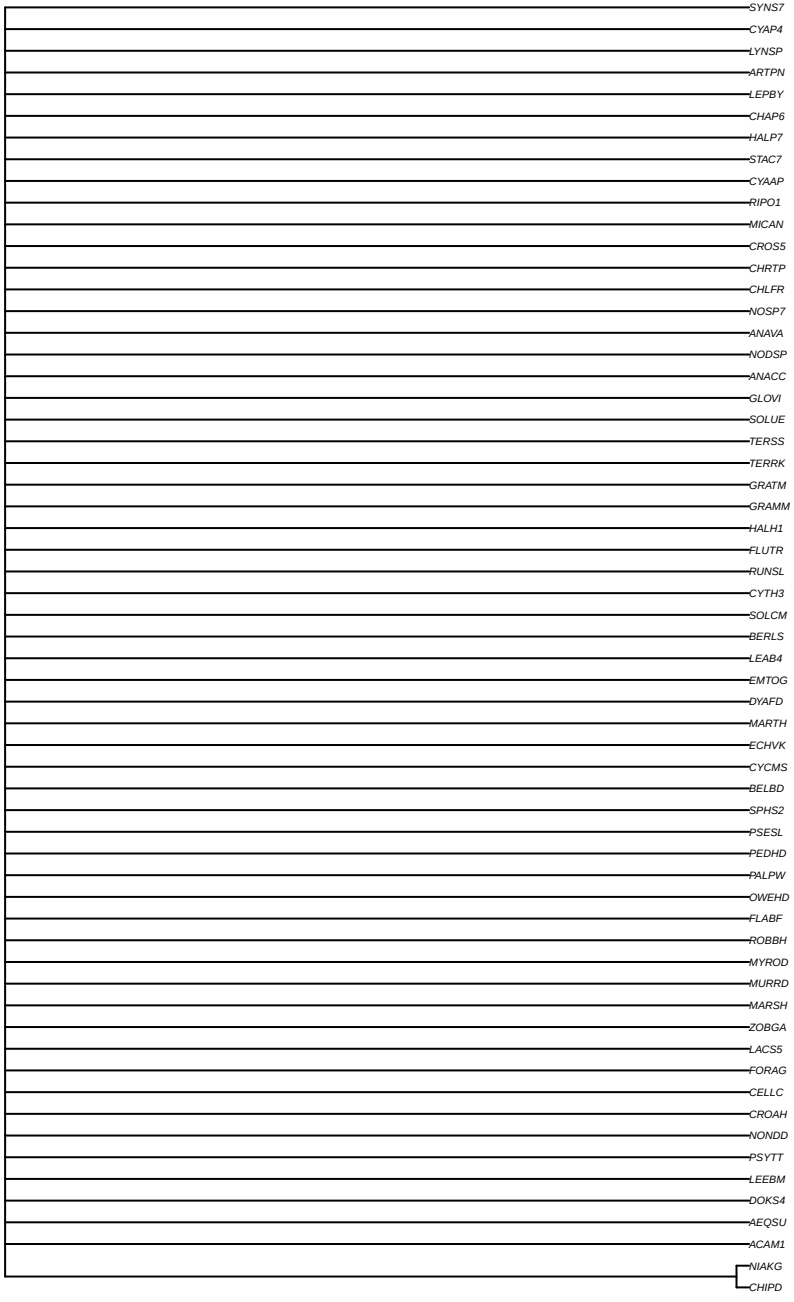

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
OB−astral
OB−ncbi−astral
OB−ncbi
MYROD
MURRD
MARSH
ZOBGA
CELLC
LACS5
FORAG
CROAH
NONDD
PSYTT
AEQSU
LEEBM
DOKS4
ROBBH
FLABF
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CHIPD
SOLUE
TERSS
GRAMM
TERRK
GRATM
GLOVI
ACAM1
SYNS7
CYAP4
LYNSP
ARTPN
LEPBY
CHAP6
HALP7
STAC7
CYAAP
RIPO1
MICAN
CROS5
CHRTP
CHLFR
NOSP7
ANAVA
NODSP
ANACC
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
MYROD
MURRD
MARSH
ZOBGA
CELLC
LACS5
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AEQSU
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GLOVI
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SYNS7
CYAP4
LYNSP
ARTPN
LEPBY
CHAP6
HALP7
STAC7
CYAAP
RIPO1
MICAN
CROS5
CHRTP
CHLFR
NOSP7
ANAVA
NODSP
ANACC
0
50
100
150
200
250
substitutions/site from root

IQ−TREE clades: number of the other 7 trees that contain each clade



strict



majority rule

